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SARDAR PATEL UNIVERSITY

BCA (Semester-IV) Examination – April 2023

[US04CBCA54] : Operating Systems

Date: 11th April 2023, Tuesday

Time: 2.00pm to 5.00pm

Total Marks: 70

Note: Figure indicate to right, is maximum marks for each question:

Q:1 Select an appropriate option for the following statement:

[10]

1. Time- Slice Scheduling algorithm is also called _____ algorithm.
(a) FCFS (b) SJF (c) Round-robin (d) Priority
2. PCB stands for _____.
(a) Program Control Block (b) Process Control Block
(c) Process Communication Block (d) Program Communication Block
3. Which one is the operating system service?
(a) Accounting (b) Error Detection (c) Communication (d) All of these
4. _____ scheduling algorithm gives minimum Page Faults.
(a) FIFO (b) Optimal (c) LRU (d) None of these
5. Which of the following memory allocation strategy is fastest?
(a) Best-fit (b) Worst-fit (c) Optimal-fit (d) First-fit
6. Demand Paging is _____.
(a) Paging with swapping (b) Paging without swapping
(c) Swapping with fragmentation (d) Swapping without fragmentation
7. _____ buffer has fixed buffer size.
(a) Bounded (b) Unbounded (c) Shared (d) None of these
8. First UNIX was developed by a _____.
(a) Linux Torvald (b) Linus Torvald (c) Ken Thompson (d) None of these
9. _____ command is used to get information about currently logged in user on to system.
(a) Who (b) login (c) syslog (d) imlogin
10. To evaluate conditional expression in Linux _____ command is used.
(a) expr (b) test (c) testexpr (d) eval

Q:2 Write brief answer for the following questions. (Attempt any Ten)

[20]

1. List types of Operating System.
2. Define: [1] Throughput [2] Context Switching
3. Draw diagram of PCB.

C.P. T. O.)

4. What is Belady's Anomaly?
5. What is Compaction? What is the use of it?
6. Explain internal fragmentation in brief.
7. When does race condition arise?
8. What is deadlock? List necessary conditions for deadlock.
9. Justify "Linux is a Secure Operating System".
10. Explain pwd command in brief.
11. Explain wildcard characters in Linux with its use.
12. Explain echo command.

Q:3 [A] What is Operating System? Explain its Services in detail. [05]

[B] Write a note on Real-Time Operating System. [05]

OR

[A] What is process? Explain process states and its transition. [05]

[B] Write a note on Multi-programming Operating System. [05]

Q:4 [A] Explain Multiprogramming with Dynamic Partitioning. [05]

[B] Write a note on Demand Paging. [05]

OR

[A] Write a note on Segmentation. [05]

[B] Discuss FIFO page replacement algorithm with example. [05]

Q:5 [A] Differentiate: Co-operative and Independent process. Also explain advantages of Co-operative process? [05]

[B] What is LINUX? Explain basic features of LINUX Operating System? [05]

OR

[A] Explain Critical-section problem in details. [05]

[B] Explain algorithm-3 for two-process solution. [05]

Q:6 [A] Explain date command with all its attribute and example. [05]

[B] Discuss case control statement with example. [05]

OR

[A] Explain wc command with its attributes. [05]

[B] List looping structure available in Linux. Explain any one of them with example. [05]

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